Несобствени интеграли

1. Интеграли от неотрицателни функции

Да се изследват за сходимост интегралите:

$$\int_{-1}^{1} e^{1/x}\frac{dx}{x^3},\quad \int_{-1}^{1}\frac{dx}{\ln(1+x)},\quad \int_{0}^{1}\frac{\ln x dx}{\sqrt{x(1-x)^3}},\quad \int_{0}^{1}\frac{\arcsin(x^2+x^3)}{x\ln^2(1+x)}dx,\quad \int_{0}^{1} e^{\alpha/x}\left(\cos x\right)^{1/x^3}dx,$$

$$\int_{0}^{1}\frac{\sqrt{e^2+x^2}-e^{\cos x}}{x^\alpha}dx,\quad \int_{0}^{\pi/2}\frac{\cos^2(2x)-e^{-4x^2}}{x^\alpha \operatorname{tg} x}dx,\quad \int_{0}^{\pi/2}\frac{e^{\alpha\cos x}-\sqrt{1+2\cos x}}{\sqrt{\cos^5 x}}dx,$$

$$\int_{0}^{1}\frac{\ln\left(e^{x^\alpha}+x\right)-x}{\operatorname{tg} x}dx,\quad \int_{1}^{\infty}\frac{1+\arcsin\frac{1}{x}}{1+x\sqrt{x}}dx,\quad \int_{0}^{\infty}\frac{\ln\left(1+x^5\right)}{\sqrt{x+\sqrt{x}}}dx,\quad \int_{0}^{\infty}\operatorname{arctg}\frac{x^3}{1+x^2}\frac{dx}{x^2},$$

$$\int_{3}^{\infty}\frac{e^{-x}-\ln x}{(1+x^\alpha)^{\alpha-2}}dx,\quad \int_{0}^{\infty}\frac{\operatorname{arctg}^{\alpha} x}{(x^2+2)\left(e^x-1\right)^\beta}dx,\quad \int_{0}^{\infty}\frac{\ln^\alpha\left(e^x-x\right)}{(x+\sqrt{x})^\beta \arcsin\frac{x}{x+1}}dx.$$

2. Интеграли от знакопроменливи функции

Да се изследват за сходимост интегралите:

$$\int_{1}^{\pi}\sin\left(\frac{1}{\cos x}\right)\frac{dx}{\sqrt{x}},\quad \int_{0}^{\infty}\frac{x\sin(5x)}{25+x^2}dx,\quad \int_{0}^{\infty}\frac{\sin\left(x+x^2\right)}{x^\alpha}dx,\quad \int_{0}^{\infty}\frac{\sin\,\ln x}{\sqrt{x}}dx$$

$$\int_{1}^{\infty}\operatorname{arctg}\frac{\cos x}{x^{2/3}}dx\quad \int_{1}^{\infty}\frac{1}{\sqrt[3]{x}}\left(\exp\left(\frac{\sin\sqrt{x}}{1+x}\right)-1\right)dx,\quad \int_{0}^{\infty}\frac{x\cos(7x)}{x^2+2x+2}dx,\quad \int_{1}^{\infty}\sin(x\ln x)dx.$$